

Ken Patrick Garcia

AI Full-Stack Engineer — RAG pipelines · AI agents · automation

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SUMMARY

Full-stack AI engineer who ships production systems end to end — RAG pipelines, LLM agents, and the automation around them. Currently building the AI automation layer for an Australian mortgage brokerage: 30 browser-agent workflows behind human approval gates, orchestrated from an email-triggered n8n pipeline, with per-run cost accounting that cut LLM spend 96%. Before that, a two-sided Flutter marketplace shipped to the App Store and Google Play, and a 31-table internal operations platform built solo. Python and TypeScript across the stack, with a bias for measuring what a system actually does.

SKILLS

Languages	Python, TypeScript, JavaScript, Java, Kotlin, SQL, HTML5, CSS3
AI / ML	RAG pipelines (pgvector · ChromaDB · sentence-transformers), AI agents (LangGraph · Vercel AI SDK · durable workflows), LLM APIs (Claude · Gemini · OpenAI · Groq · OpenRouter), local inference (Ollama · Gemma · YOLO → TFLite), Rasa, n8n Automation (AI Agent orchestrators · tool calling), browser agents (Browser Use · Skyvern · Playwright-over-CDP), agentic workflows with human-in-the-loop gates, LLM cost engineering & model routing, multi-model benchmarking, structured output (JSON Schema), TensorFlow, Scikit-learn, Computer Vision, NLP, Recommendation Systems
Frontend & Mobile	React.js, Next.js, React Native (Expo), Flutter, Riverpod, GoRouter, Astro, Three.js, TailwindCSS, Zustand, Recharts, Motion, Responsive Design
Backend & DB	Node.js, Spring Boot, FastAPI, PostgreSQL (pgvector · RLS), Supabase (RLS · column grants · SECURITY DEFINER RPCs · Realtime · Storage), Supabase Edge Functions (Deno), PostgREST, Firebase (Auth · Firestore · Hosting · Cloud Messaging), Drizzle ORM, Prisma ORM, Redis, Upstash Redis, RESTful APIs, Socket.io
DevOps & Cloud	AWS (EC2 · S3 · VPC · EBS), GCP (Cloud Run · Gemini API — Google Gen AI Academy APAC), Docker, Kubernetes (learning), GitHub Actions, CI/CD Pipelines, Vercel, Coolify, Cloudflare Tunnel, Sentry, Browserbase, Browser Use Cloud, Steel (self-hosted), residential proxy pinning
Testing & QA	Vitest, Playwright E2E, live-database audit pipelines, mutation testing, static checkers as CI gates
Specialized	IoT Integration, Real-time Data Processing, WebSockets, Performance Optimization, Human-in-the-loop Approval Architecture, Irreversible-action Safety Design, IMAP / TOTP 2FA Automation, Audit Trails for Regulated Financial Workflows, Offline-first Caching, App Store & Google Play Release Management, GDPR Compliance
AI Workflow	AI-native development — Claude Code on Claude Max (daily agentic coding, custom skills & automation), ChatGPT Pro (research & prototyping), self-hosted Ollama + Open WebUI for local models

EXPERIENCE

Mufti Finance · AI Automation Engineer (Contract)

Jul 2026 – Present · Australia (Remote)

- Built the AI automation layer for an Australian mortgage brokerage: 30 browser-agent workflows across six major Australian lender portals that log in, complete multi-step valuation and pricing wizards, and stop at a human approval gate — ~18K lines of Python plus n8n orchestrators and two Next.js operator consoles.
- Designed an email-triggered orchestration pipeline in n8n: an inbound broker email is classified by an LLM agent, missing fields are gathered conversationally over reply threads, PDF attachments are uploaded to object storage and mapped to the correct form fields, and the run is dispatched to a browser agent — all 9 request types proven end-to-end, email to live lender portal in ~70–95s.
- Cut per-run LLM cost 96% (\$0.98 → \$0.04) with a cheap-first model router that escalates only on flow-specific completion markers, and replaced agent-driven logins with Playwright-over-CDP scripted logins (87s, \$0.00, zero tokens) — measured against a token-accounting harness built into every run.
- Shipped the safety architecture the automation runs behind: human approval gates on every irreversible step, AU residential proxy pinning, claim-before-dispatch to prevent duplicate orders, and a verifier that reads each workflow's live definition rather than its description — which caught 10 of 19 workflows whose safety flags contradicted their own code.

Jobdun Pty Ltd · Lead Full-Stack & Mobile Engineer (Contract)

May 2026 – Aug 2026 · Sydney, Australia (Remote)

- Built and shipped Jobdun, a two-sided job marketplace for the Australian construction trades, from empty repo to live on the Apple App Store (AU) and Google Play — ~70K lines of Dart across 566 files, one Flutter codebase serving iOS, Android, and web.
- Designed the Supabase backend end to end: 32 Postgres tables under row-level security, 88 migrations, and 5 Deno edge functions covering the jobs feed, push delivery, and ABN / trade-licence verification against Australian regulator APIs.
- Cut jobs-feed latency with an Upstash Redis read-through cache (45s TTL, write invalidation, kill switch) in front of the feed edge function, backed by an encrypted on-device Hive cache for stale-while-revalidate reads and offline browsing.
- Shipped the platform around the app: a Next.js admin console on Vercel with an SSR admin-role gate (verification queue, user/job moderation, audit feed, broadcast), the marketing site at jobdun.com.au, FCM push on both platforms, Twilio SMS OTP, Google/Apple sign-in, Sentry, and a GitHub Actions + pre-push gate enforcing format, lint, tests, and Clean Architecture boundaries.

Romega Solutions · AI Full-Stack Engineer (Part-Time)

Jul 2025 – Jul 2026 · El Segundo, CA (Remote)

- Built an all-in-one internal operations platform (PM/ticketing, attendance, recruiting ATS, LMS, AI reporting) serving a distributed team — 46 pages, 56 API routes, and a 31-table Postgres schema in ~42K lines of TypeScript, shipped solo in ~3 months.
- Led a zero-downtime migration off a third-party PM SaaS: rebuilt ticketing natively (kanban, sprints, comments, activity feeds), migrated SQLite → Supabase Postgres with Drizzle ORM, and replaced custom JWT auth with Supabase Auth + Google OAuth.
- Integrated LLM features (daily executive briefings, AI status-report drafting) and n8n automations (resume parsing, applicant comms, multi-step onboarding sequences) directly into business workflows — outputs land in human review queues, never auto-publish.
- Shipped production RAG chatbots with vector embeddings (ChromaDB · sentence-transformers) and Google Gemini for customer interactions and workflow automation.

CodeVF · Software Engineer (Contract)

Feb 2026 – Apr 2026 · US (Remote)

- Engineered a real-time collaboration platform with WebSocket architecture, enabling live debugging sessions between customers and engineers with bidirectional communication for code review and remote assistance.

University of Makati · AI Workflow Automation Engineer (Part-Time)

Jan 2026 — Apr 2026 · Makati (Remote)

- Developed intelligent automation workflows using n8n to streamline university administrative processes and data management.

Pacebeats · Lead Software Engineer (Full-Time)

Aug 2025 — Mar 2026 · Makati (Remote)

- Architected a biometric music platform (1st Runner-Up, InfoTech Olympics 2025) with a WearOS app (Kotlin, Health Services API), Kotlin Android app, React/TypeScript dashboard, and Supabase backend.
- Built a hybrid recommendation engine combining rule-based pace-to-BPM mapping with content-based ML scoring on user listening patterns, achieving real-time playlist adaptation.
- Engineered the WearOS companion with GPS tracking, heart-rate monitoring, Data Layer API sync, Spotify SDK integration, and 8+ hour battery optimization.

Commission on Elections (COMELEC) · Technical DESO Support Staff (Contract)

Jul 2025 — Oct 2025 · Makati

- Installed and configured Vote Counting Machine (VCM) hardware — CF/SD cards, thermal printers, modems, batteries — for Final Testing & Sealing and Election Day operations.

Concentrix · Technical Support Representative

Feb 2024 — Sep 2024 · Mandaluyong

- Provided technical support for a major enterprise SaaS productivity platform (client under NDA), troubleshooting complex issues across identity management, cloud services, and software configuration.

EDUCATION

University of Makati · BS Computer Science, Major in Application Development

Makati City

Technological Institute of the Philippines · BS Computer Science

2020 — 2021 · Manila

AWARDS & HONORS

- 1st Runner-Up** — InfoTech Olympics 2025 (University of Makati), Android App Development: Productivity (Pacebeats)
- Top 10 Finalist** — DOST-TAPI Regional Invention Contest 2025, 53 entries (HerbaLens)
- Best Paper Presentation** — 8th Research Congress, University of Makati (ARS)
- Champion** — C(Old) (St)art Hackathon 2025 by Old St. Labs

LEADERSHIP & COMMUNITY

- Speaker** — Qwen Meetup Manila #2 (Alibaba Cloud PH), 2026
- DataCamp Scholar** — Data Engineering Pilipinas, Nov 2025 – Present
- AWS Community Officer** — AWS User Group PH, Community Day 2026
- Technical Committee** — UMak Computer Society, 15+ campus events

LANGUAGES

Filipino (Native) · English (Professional)